proPlan

Intelligent University Course Schedule Planner

Web Application Project Proposal • Fall 2026 • Project 1

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Repository https://github.com/mirayyazici/SOFT3111-proPlan.git

2.2 Project Overview and Motivation

University course registration is often stressful and time-consuming. Students must compare semester timetables, course sections, prerequisites and credit limits while considering failed or conditionally passed courses. They must also understand how their choices affect graduation progress. Relevant information is scattered across transcripts, degree curricula, registration regulations and semester course lists. Manually creating a valid weekly schedule becomes especially difficult when compulsory classes compete for limited time slots or students are enrolled in double-major or minor programs.

proPlan is an AI-assisted web application designed initially for Işık University students. Students register or sign in using their Işık University-issued email address, then review their academic profile. A student portal (e-Campus) integration is planned to retrieve an authenticated student's transcript, course attempts, GPA, credit eligibility and available CAR/CCR information, but no authorized live API is currently available. Therefore, the first version will use clearly labeled mock academic records and permit students to upload transcript PDFs. An administrator will upload the semester course schedule PDF and verify extracted course sections, credits, days and meeting times.

A deterministic scheduling engine, developed in C# with ASP.NET Core, will apply the relevant Işık University academic policies: course and earned-credit prerequisites, mandatory repeats, GPA-based ECTS limits, curriculum sequence, double-major (ÇAP) and minor (YAP) requirements, and section availability. It will prioritize clash-free schedules and flag any potentially permitted one-teaching-period overlap for adviser review rather than treating it as automatically approved. Students may choose a lower credit target and compare multiple feasible alternatives. The weekly calendar follows 50-minute lessons starting at 08:30, with ten-minute breaks.

An AI assistant will translate natural-language preferences, such as keeping Mondays free or selecting artificial-intelligence electives, into structured filters and recommendations from offered, eligible courses. AI will not override the academic rule engine. The first release focuses solely on Işık University; support for additional universities may be developed later through separate academic-policy, email-domain, curriculum and authorized portal integrations. This staged approach keeps the semester project realistic while allowing future expansion.

Initial Scope and Future Expansion

Initial institution: Işık University only. The initial sign-in and sign-up screens do not provide a university selector. Işık University is the configured pilot institution, and the application validates the institutional student email domain. Having a matching email domain alone does not establish student identity. An actual e-Campus request requires university authorization, secure authentication and student consent. The first prototype uses mock records and optional student-uploaded transcripts. Actual course enrollment, adviser approval and live seat availability are outside the first version. Other universities are a possible future enhancement, not part of the initial deliverable.

2.3 Target Users and User Roles

Role	Expectations	Permitted actions
Student	Plan an academically eligible Işık University semester without manually checking every restriction.	Register/sign in with verified Işık University email; inspect mock or authorized own academic data; optionally upload transcript; choose course load and preferences; compare and save plans.
Authorized Administrator	Maintain a validated Işık University semester offering dataset and policy settings.	Upload and verify semester PDFs; correct course-section and lab information, credits, time slots and offered capacities; publish reviewed data for planning.

Application Navigation

Initial navigation: Welcome, Işık University sign-up/sign-in, Student Portal (e-Campus demo), transcript import, academic dashboard, AI preferences, generated schedules, comparison and semester administration. For the first release, Işık University is fixed in the application configuration and no university-selection field appears; multi-university selection and other portal connectors are possible future extensions. The browser prototype demonstrates navigation and sample validation; production login and private records require an authorized ASP.NET Core service.

2.4 Functional Requirements

ID	Testable requirement	Priority
FR-01	Students can register and sign in with a verified Işık University email. The pilot has no university selector; an email-domain match or Student ID alone does not verify identity or grant access to private records.	High
FR-02	Authorized administrators can upload semester schedule PDFs, preview extracted course codes, titles, sections, ECTS, days and required lecture/lab meetings, correct errors and publish reviewed offerings.	High
FR-03	Students can review transcript, course history, grades, GPA, academic status, credit eligibility, remaining requirements and CAR/CCR when available. Until an authorized e-Campus API exists, portal data is labeled DEMO; transcript PDF upload is the fallback. Real records require secure authentication, university authorization and student consent.	High
FR-04	The C# engine validates course and earned-ECTS prerequisites, curriculum sequence and first-year exceptions using the policy version applicable to the admission cohort; upper-semester selections requiring approval are flagged.	High
FR-05	The engine uses GPA and historical warning/repeating status to identify mandatory F repeats and applicable DD/DC obligations, prioritizing required repeats that are offered.	High
FR-06	The system calculates the student-specific maximum ECTS from academic status and permits a chosen target at or below that maximum.	High
FR-07	Students can select Normal, ÇAP or YAP separately, never combined ÇAP+YAP. The engine checks approved secondary curricula and equivalencies without double-counting ECTS, applies pathway-specific ceilings and restricts additional ECTS to eligible secondary-program courses.	High
FR-08	The engine prioritizes clash-free plans. Where policy permits, only one exceptional clash involving two courses for one teaching period is flagged for adviser review, never automatically approved.	High
FR-09	The system generates up to five distinct academically feasible weekly plans from offered sections when enough combinations exist; otherwise it explains why fewer or no plans are available.	High
FR-10	Students can set free days, earliest/latest class times, target credits, campus-day preferences and elective interests.	High
FR-11	The timetable displays 50-minute lessons starting at 08:30, with ten-minute breaks. Section capacities are recorded only from reviewed data; missing or outdated quotas are marked unverified.	High
FR-12	Students can navigate through Welcome, Işık University Sign In/Sign Up, Student Portal, Transcript Import, Dashboard, AI Preferences, Results and Comparison.	High
FR-13	Generated plans are labeled advisory. Actual registration, quota exceptions and final approval remain with the university and adviser.	High
FR-14	The AI assistant converts natural-language requests into validated structured preferences and recommends only offered, academically eligible electives, explaining their relevance to stated interests. AI cannot override academic rules.	Medium

ID	Testable requirement	Priority
FR-15	Students can compare weekly plans and save a preferred plan without enrolling in courses.	Medium
FR-16	The interface explains unmet prerequisites, quota constraints, curriculum-sequence restrictions, unsatisfied preferences and adviser-review cases.	Medium

2.5 Non Functional Requirements

ID	Measurable requirement	Quality
NFR-01	Desktop pages and timetable views work at 1440 px, and essential controls remain usable at 375 px mobile width without clipping core content.	Usability
NFR-02	Only authenticated, verified university accounts may access private records. Student ID alone gives no access, and secrets or real transcripts are not placed in the public GitHub repository.	Security
NFR-03	For a fixed sample catalog and student record, the scheduling service returns up to five options within 10 seconds on the project development machine.	Performance
NFR-04	Invalid PDFs, unavailable portal data, missing prerequisite information and infeasible schedules produce understandable messages without application crashes.	Reliability
NFR-05	Main forms, buttons and navigation links are keyboard accessible, with readable labels and visible focus indicators.	Accessibility
NFR-06	The main user flows will be tested on recent Chrome and Firefox versions using C# Selenium WebDriver, with separate unit tests for academic rules.	Compatibility

2.6 User Scenarios

Scenario 1 Import Semester Data

Actor Authorized Administrator

Preconditions An authorized administrator and an Işık University semester course PDF are available.

1. Sign in as administrator.
2. Upload the semester PDF.
3. Review extracted codes, sections, lecture/lab meetings and ECTS.
4. Correct uncertain values.
5. Publish the verified semester dataset.

Expected outcome Only reviewed Işık University offerings are used for scheduling. (FR-02)

Scenario 2 Build a Valid Schedule

Actor Işık University Student

Preconditions A reviewed semester dataset and an authenticated or clearly labeled mock academic profile exist.

1. Sign in with an Işık University school email.
2. Enter Student ID and load demo portal data, or upload a transcript.
3. Review the academic record and retake requirements.
4. Select a permitted target credit load and free-day preferences.
5. Compare generated weekly plans.

Expected outcome Plans respect the student-specific ECTS maximum, valid offered sections and academic rules; unresolvable exceptions are explained. (FR-01, FR-03–FR-06, FR-08–FR-10, FR-16)

Scenario 3 Conversational Personalization

Actor Işık University Student

Preconditions An academic profile and approved semester offerings are available.

1. Open AI Preferences.
2. Type, “Keep Monday free and prefer AI electives.”
3. Review the extracted preferences.
4. Generate new plans.
5. Compare alternatives.

Expected outcome Monday-free options are ranked first when feasible, and electives come only from available eligible courses. (FR-10, FR-14, FR-15)

Scenario 4 Resolve an Academic Constraint

Actor Işık University Student

Preconditions The record includes a failed compulsory course currently offered by the university.

1. Request a plan conflicting with a mandatory repeat.
2. The rule engine checks prerequisites, repeated courses and teaching-period clashes.
3. Review warnings and any adviser-dependent exception.
4. Change the requested load or free-day preference.

Expected outcome Mandatory academic rules remain enforced; a possible one-period clash is never treated as automatic enrollment approval. (FR-04, FR-05, FR-08, FR-16)

Scenario 5 Double Major and Minor Planning

Actor Işık University ÇAP or YAP Student

Preconditions An approved secondary-program record and equivalency information are available.

1. Select ÇAP or YAP (not both).
2. Review the main and secondary curricula.
3. Check equivalent courses and the track-specific ECTS limit.
4. Select a target load.
5. Generate and compare alternatives.

Expected outcome The applicable secondary-program requirements and extra-credit restrictions are enforced without double-counting equivalent courses. (FR-07)

Scenario 6 Işık University Sign-In and Portal Data

Actor Işık University Student

Preconditions The student has an institution-issued email and an account for the prototype.

1. Open Sign In/Sign Up without a university-selection control; the pilot institution is Işık University.
2. Enter a valid school email.
3. After demo login, open Student Portal.
4. Enter Student ID and choose sample data or transcript upload.
5. Review GPA, CAR/CCR and transcript fields.

Expected outcome The student can complete the demo flow, with mocked data clearly labeled. Authorized e-Campus retrieval remains future integration work. (FR-01, FR-03, FR-12)